



JADARD

JD79661AA

Customer User Guide

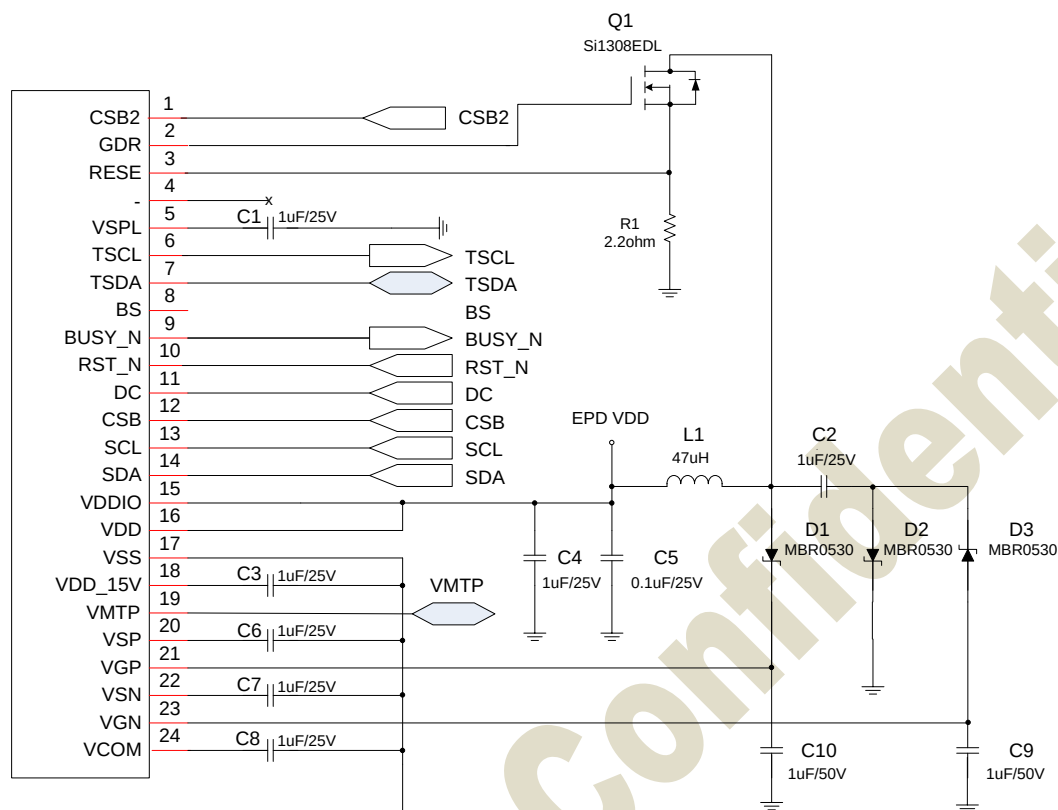
All-in-one Driver with
TCON for Color Application

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All-in-One Driver with TCON for Color Application

1. APPLICATION CIRCUIT



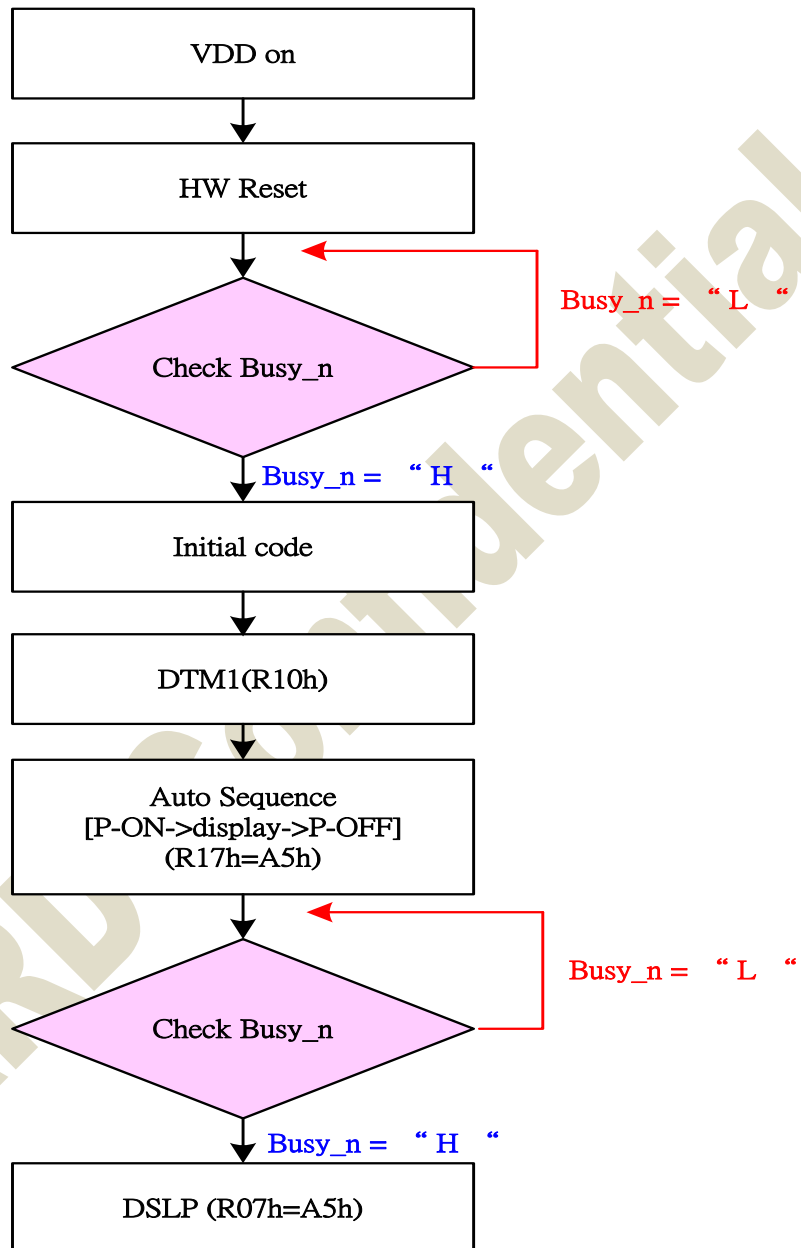
Device no.	Value	Reference
C1,C2,C3, C4, C6, C7, C8	1uF	0603, X5R/X7R, voltage rating : 25V
C9, C10	1uF	0603, X5R/X7R, voltage rating : 50V
C5	0.1uF	0603, X5R/X7R, voltage rating : 25V
R1	2.2Ω	0603, +/-1% variation
Q1	NMOS	Si1308EDL、Si1304BDL - Drain-source break volatage $\geq 30V$ - Gate-source threshold voltage $\leq 1.5V$ - Drain-source on-state resistance $< 400m\Omega$
L1	47uH	NR4018T470M、CDRH2D18/LDNP-470NC - Fixed - Maximum DC current $\sim 420mA$ - Maximum DC resistance $\sim 650m\Omega$
D1~D3	Diode	MBR0530 - Reverse DC voltage $\geq 30V$ - Forward current $\geq 500mA$ - Forward voltage $\leq 430mV$

Note:

1. Power board 可共用新的及其他 compatible IC 的應用電路，搭配不同的外部電部會有相應的 boost(R06) 參數需調整
2. VSP(C6)/VSN(C7)/VSPL(C1) 電容不可更換其他容值，會影響 mode change 時，電壓爬升時間

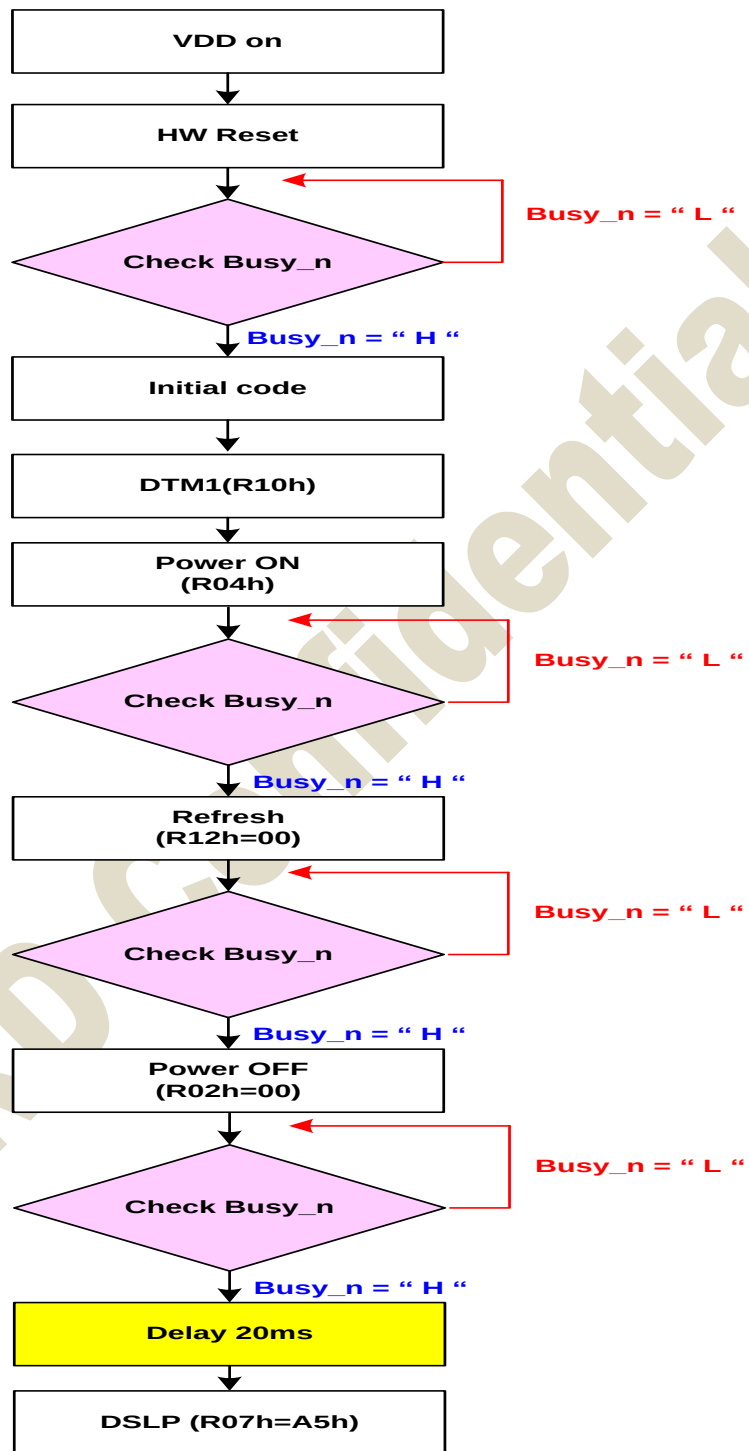
2. DISPLAY FLOW

2.1 Display Flow A



Note: MTP 燒錄後的模組 display flow 不需再寫 LUT

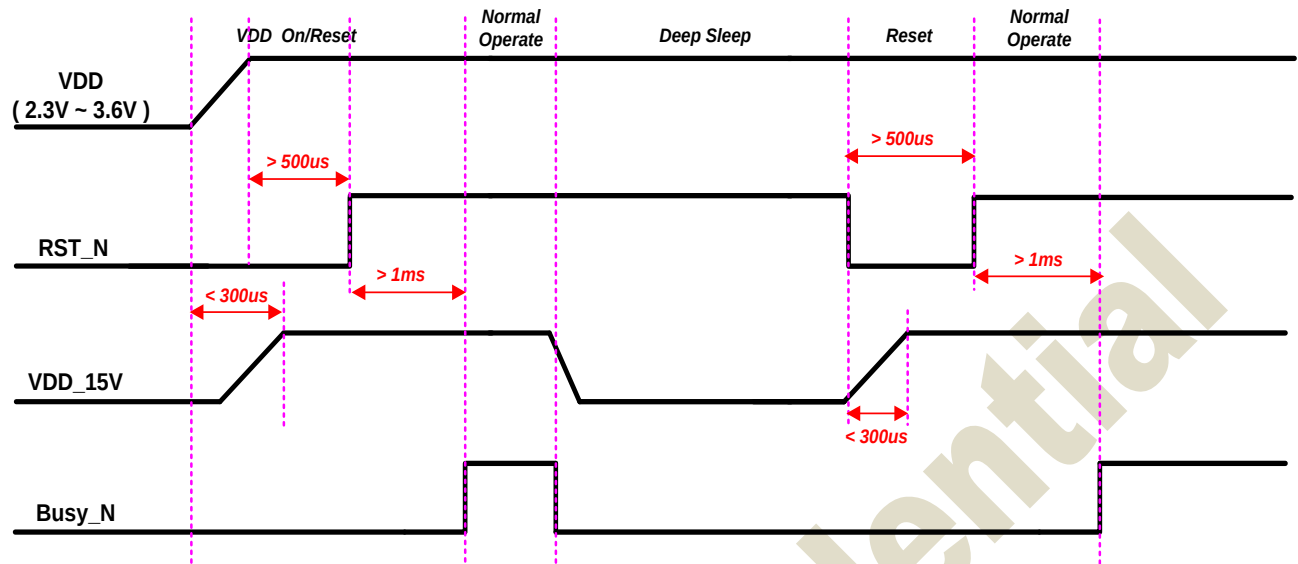
2.2 Display Flow B



Note:

1.MTP 燒錄後的模組 display flow 不需再寫 LUT

2.3 HW Reset



Note:

1. 重新上電後，務必作 hw reset(low->high)來作重置，確認 IC 回到最初狀態
2. Reset 後需偵測 busy_n 拉 high，來確定此時 IC 重置完成進至 normal 狀態，可以再進行其他動作
3. 進入 deep sleep mode 後，digital 電壓 VDD_15V 已關無法在下 command，如需持續操作要透過外部 hw reset(RST_N : low->high)來喚醒

2.4 Initial Code

2.4.1 Before MTP Model

Description	Address(Hex)	Data(Hex)							Note
PWD	4D	78	-	-	-			-	
	B4	D0	-	-	-			-	
User Cmd	03	10	54	44	-			-	
	06	0F	0A	2F	25	22	2E	21	
	61	00	B0	01	08			-	176S x 264G resolution setting
	E3	22	-	-	-			-	

2.4.2 After MTP Model

Description	Address(Hex)	Data(Hex)
PWD	4D	78
	B4	D0
	AE	0F
	B6	6F

Note:

- 1.上述 user command 的 data 為參考值，需依實際模組狀況調整
- 2.各面板解析度參考設定如下：

Description	Address(Hex)	Data(Hex)				Resolution
Resolution Setting	61	00	68	00	D4	104S x 212G
	61	00	98	00	98	152S x 152G
	61	00	90	00	C8	144S x 200G
	61	00	80	00	FA	128S x 250G
	61	00	80	01	28	128S x 296G
	61	00	98	01	28	152S x 296G
	61	00	B0	01	08	176S x 264G

Resolution setting for panel application

2.5 Busy_N Flag

部份 cmd. 執行時 busy_n 會拉 low 則 IC 進入工作狀態，工作結束時會再拉回 high；建議執行會 flag 的 cmd. 時，MCU 需等 busy_n 拉 high 後再執行其他工作。

Register	Refresh Restriction	BUSY_N flag
R00H(PSR)	X	No action
R01H(PWR)	X	No action
R02H(POF)	X	Flag
R03H(PFS)	X	No action
R04H(PON)	X	Flag
R06H(BTST)	X	No action
R07H(DSLP)	X	Flag
R10H(DTM1)	X	No action
R11H(DSP)	Valid only read	Flag
R12H(DRF)	X	Flag
R17H(AUTO)	Valid in standby	Flag
R30H(PLL)	X	No action
R40H(TSC)	Valid only read	Flag
R41H(TSE)	X	No action
R42H(TSW)	X	Flag
R43H(TSR)	Valid only read	Flag
R50H(CDI)	X	No action
R51H(LPD)	Valid only read	Flag
R61H(TRES)	X	No action
R65H(GSST)	X	No action
R70H(REV)	Valid only read	No action
R80H(AMV)	X	Flag
R81H(VV)	Valid	No action
R82H(VDCS)	X	No action
R83H(PTL)	X	No action
R90H(PGM)	X	No action
R91H(APG)	X	Flag
R92H(RMTP)	X	Flag
R9EH(REV2)	X	No action
R9FH(Read MTP reserved)	Valid only read	Flag
RE3H(PWS)	X	No action
RE4H(LVSEL)	X	No action

3. REGISTER DESCRIPTION

D/CX:0:Command/1:Data

3.1 User Command

3.1.1 R00H(PSR): Panel Setting Register

R00H	Bit										
Inst/Para	R/W	D/CX	D7	D6	D5	D4	D3	D2	D1	D0	Code
PSR	W	0	0	0	0	0	0	0	0	0	00H
1 st Parameter	W	1	RES[1]	RES[0]	PST_MODE	-	UD	SHL	SHD_N	RST_N	0Fh
2 nd Parameter	W	1	LUT_EN	-	FOPT	VCMZ	TS_AUTO	TIEG	NORG	VC_LUTZ	09h

NOTE: “-” Don't care, can be set to VDD or GND level

Description	- The command defines as :		
	1st parameter		
	Bit	Name	Description
	0	RST_N	RST_N function 1: no effect. (default) 0: Booster OFF, Register data are set to their default values, and Source/Boder/Vcom: floating
	1	SHD_N	SHD_N function 0 : Booster OFF, register data are kept, and Source/Boder/Vcom are kept 0V or floating. 1 : Booster on. (default)
	2	SHL	SHL function 0: Shift left; First data=Sn→Sn-1 →...→S2→Last data=S1. 1: Shift right: First data=S1→S2 →...→Sn-1→Last data=Sn. (default)
	3	UD	UD function 0:Scan down; First line=Gn→Gn-1 →...→G2→Last line=G1. 1:Scan up; First line=G1→G2 →...→Gn-1→Last line=Gn. (default)
	5	PST_MODE	Power switch operation mode 0:Power switching time in the period of frame scanning.(default) 1:Power switching time in the external period before frame scanning.
	7-6	RES[1,0]	Resolution setting 00: Display resolution is 176x296 (default) 01: Display resolution is 128x296 10: Display resolution is 128x250 11: Display resolution is 112x204
	0	RST_N	RST_N function 1: no effect. (default) 0: Booster OFF, Register data are set to their default values, and Source/Boder/Vcom: floating

	2 nd parameter		
	Bit	Name	Description
	0	VC_LUTZ	VCOM status function 0 : No effect 1 : After refreshing display,the output of VCOM is set to floating automatically (default)
	1	NORG	VCOM status function 0 : No effect (default) 1 : After refreshing display, VCOM is tied to GND before power off
	2	TIEG	VGN power off status function 0 : No effect (default) 1 : Power off, VGN will be tied to GND
	3	TS_AUTO	Temperature sensing will be activated automatically one time 0 : Before enabling booster, Temperature Sensor will be activated automatically one time. 1 : When RST_N low to high, Temperature Sensor will be activated automatically one time. (default)
	4	VCMZ	VCOM status function 0 : No effect (default) 1 : VCOM is always floating
	5	FOPT	FOPT function 0: Scan 1 frame after waveform finished(default) 1: No scan after waveform finished and switch the source channel output to Hiz.
	Priority of VCOM setting: VCMZ > NORG > FOPT > VC_LUTZ FOPT setting is part of refreshing display. FOPT: Power off floating. Notes: 1. Non-select gate line keep at VGN for DSP/DRF and AMV 2. Dummy source line follow LUTC for DSP/DRF 3. When SHD_N become low, DCDC will turn off. Register and SRAM data will keep until VDD turn off. SD output and VCOM will base on previous condition. It may have two condition: 0V or floating. 4. When RST_N become low, driver will reset. All register will reset to default value. All of the driver's functions will disable. Source/Gate/Border/VCOM will be released to floating		
Restriction			

3.1.2 R01H(PWR): Power Setting Register

R01H	Bit										
Inst/Para	R/W	D/CX	D7	D6	D5	D4	D3	D2	D1	D0	Code
PWR	W	0	0	0	0	0	0	0	0	1	01h
1 st Parameter	W	1	-	-	-	-	V_MODE	VSC_EN	VDS_EN	VDG_EN	07h
2 nd Parameter	W	1	-	-	-	-	-	-	VGPN [1]	VGPN [0]	00h
3 rd Parameter	W	1	-	VSPL_0 [6:0]							00h
4 th Parameter	W	1	-	VSP_1 [6:0]							00h
5 th Parameter	W	1	-	VSN_1 [6:0]							00h
6 th Parameter	W	1	-	VSPL_1 [6:0]							00h

NOTE: “-” Don’t care, can be set to VDD or GND level

Description -The command defines as :

1st Parameter:

Bit	Name	Description
0	VDG_EN	Gate power selection. 0 : External gate power from VGP/VGN pins. 1 : Internal DCDC function for generate VGP/VGN. (default)
1	VDS_EN	Source power selection. 0 : External source power from VSP/VSN pins. 1 : Internal regulator function for generate VSP/VSN (default)
2	VSC_EN	Source LV power selection. 0 : External source power from VSPL pins. 1 : Internal regulator function for generate VSPL (default)
3	V_MODE	Source Power switching mode. 0: Mode0(default) 1: Mode1

2nd Parameter:

Bit	Name	Description
1-0	VGPN	VGPN Voltage Level. 00: VGP=20 v, VGN=-20v (default) 01: VGP=17 v, VGN=-17v 10: VGP=15 v, VGN=-15v 11: VGP=10 v, VGN=-10v

3rd & 4th & 6th Parameter: Internal VSP_1/VSPL_0/ VSPL_1 power selection

Bit	Name	Description											
6-0	VSP_1 & VSPL_0 & VSPL_1	Internal VSP & VSPL power selection.											
		bit[6:0]		Voltage(V)		bit [6:0]		Voltage(V)		bit [6:0]		Voltage(V)	
		0000000	00h	3	0101001	29h	7.1	1010010	52h	11.2			
		0000001	01h	3.1	0101010	2Ah	7.2	1010011	53h	11.3			
		0000010	02h	3.2	0101011	2Bh	7.3	1010100	54h	11.4			
		0000011	03h	3.3	0101100	2Ch	7.4	1010101	55h	11.5			
		0000100	04h	3.4	0101101	2Dh	7.5	1010110	56h	11.6			
		0000101	05h	3.5	0101110	2Eh	7.6	1010111	57h	11.7			
		0000110	06h	3.6	0101111	2Fh	7.7	1011000	58h	11.8			
		0000111	07h	3.7	0110000	30h	7.8	1011001	59h	11.9			
		0001000	08h	3.8	0110001	31h	7.9	1011010	5Ah	12			
		0001001	09h	3.9	0110010	32h	8	1011011	5Bh	12.1			
		0001010	0Ah	4	0110011	33h	8.1	1011100	5Ch	12.2			
		0001011	0Bh	4.1	0110100	34h	8.2	1011101	5Dh	12.3			
		0001100	0Ch	4.2	0110101	35h	8.3	1011110	5Eh	12.4			
		0001101	0Dh	4.3	0110110	36h	8.4	1011111	5Fh	12.5			
		0001110	0Eh	4.4	0110111	37h	8.5	1100000	60h	12.6			
		0001111	0Fh	4.5	0111000	38h	8.6	1100001	61h	12.7			
		0010000	10h	4.6	0111001	39h	8.7	1100010	62h	12.8			
		0010001	11h	4.7	0111010	3Ah	8.8	1100011	63h	12.9			
		0010010	12h	4.8	0111011	3Bh	8.9	1100100	64h	13			
		0010011	13h	4.9	0111100	3Ch	9	1100101	65h	13.1			
		0010100	14h	5	0111101	3Dh	9.1	1100110	66h	13.2			
		0010101	15h	5.1	0111110	3Eh	9.2	1100111	67h	13.3			
		0010110	16h	5.2	0111111	3Fh	9.3	1101000	68h	13.4			
		0010111	17h	5.3	1000000	40h	9.4	1101001	69h	13.5			
		0011000	18h	5.4	1000001	41h	9.5	1101010	6Ah	13.6			
		0011001	19h	5.5	1000010	42h	9.6	1101011	6Bh	13.7			
		0011010	1Ah	5.6	1000011	43h	9.7	1101100	6Ch	13.8			
		0011011	1Bh	5.7	1000100	44h	9.8	1101101	6Dh	13.9			
		0011100	1Ch	5.8	1000101	45h	9.9	1101110	6Eh	14			
		0011101	1Dh	5.9	1000110	46h	10	1101111	6Fh	14.1			
		0011110	1Eh	6	1000111	47h	10.1	1110000	70h	14.2			
		0011111	1Fh	6.1	1001000	48h	10.2	1110001	71h	14.3			
		0100000	20h	6.2	1001001	49h	10.3	1110010	72h	14.4			
		0100001	21h	6.3	1001010	4Ah	10.4	1110011	73h	14.5			
		0100010	22h	6.4	1001011	4Bh	10.5	1110100	74h	14.6			
		0100011	23h	6.5	1001100	4Ch	10.6	1110101	75h	14.7			
		0100100	24h	6.6	1001101	4Dh	10.7	1110110	76h	14.8			
		0100101	25h	6.7	1001110	4Eh	10.8	1110111	77h	14.9			
		0100110	26h	6.8	1001111	4Fh	10.9	1111000	78h	15			
		0100111	27h	6.9	1010000	50h	11	other		15			
		0101000	28h	7	1010001	51h	11.1						

5th Parameter: Internal VSN_1 power selection

Bit	Name	Description									
6-0	VSN_1	Internal VSN power selection.									
		bit[6:0]	Voltage(V)	bit [6:0]	Voltage(V)	bit [6:0]	Voltage(V)	bit [6:0]	Voltage(V)	bit [6:0]	Voltage(V)
		0000000	00h	-3	0101001	29h	-7.1	1010010	52h	-11.2	
		0000001	01h	-3.1	0101010	2Ah	-7.2	1010011	53h	-11.3	
		0000010	02h	-3.2	0101011	2Bh	-7.3	1010100	54h	-11.4	
		0000011	03h	-3.3	0101100	2Ch	-7.4	1010101	55h	-11.5	
		0000100	04h	-3.4	0101101	2Dh	-7.5	1010110	56h	-11.6	
		0000101	05h	-3.5	0101110	2Eh	-7.6	1010111	57h	-11.7	
		0000110	06h	-3.6	0101111	2Fh	-7.7	1011000	58h	-11.8	
		0000111	07h	-3.7	0110000	30h	-7.8	1011001	59h	-11.9	
		0001000	08h	-3.8	0110001	31h	-7.9	1011010	5Ah	-12	
		0001001	09h	-3.9	0110010	32h	-8	1011011	5Bh	-12.1	
		0001010	0Ah	-4	0110011	33h	-8.1	1011100	5Ch	-12.2	
		0001011	0Bh	-4.1	0110100	34h	-8.2	1011101	5Dh	-12.3	
		0001100	0Ch	-4.2	0110101	35h	-8.3	1011110	5Eh	-12.4	
		0001101	0Dh	-4.3	0110110	36h	-8.4	1011111	5Fh	-12.5	
		0001110	0Eh	-4.4	0110111	37h	-8.5	1100000	60h	-12.6	
		0001111	0Fh	-4.5	0111000	38h	-8.6	1100001	61h	-12.7	
		0010000	10h	-4.6	0111001	39h	-8.7	1100010	62h	-12.8	
		0010001	11h	-4.7	0111010	3Ah	-8.8	1100011	63h	-12.9	
		0010010	12h	-4.8	0111011	3Bh	-8.9	1100100	64h	-13	
		0010011	13h	-4.9	0111100	3Ch	-9	1100101	65h	-13.1	
		0010100	14h	-5	0111101	3Dh	-9.1	1100110	66h	-13.2	
		0010101	15h	-5.1	0111110	3Eh	-9.2	1100111	67h	-13.3	
		0010110	16h	-5.2	0111111	3Fh	-9.3	1101000	68h	-13.4	
		0010111	17h	-5.3	1000000	40h	-9.4	1101001	69h	-13.5	
		0011000	18h	-5.4	1000001	41h	-9.5	1101010	6Ah	-13.6	
		0011001	19h	-5.5	1000010	42h	-9.6	1101011	6Bh	-13.7	
		0011010	1Ah	-5.6	1000011	43h	-9.7	1101100	6Ch	-13.8	
		0011011	1Bh	-5.7	1000100	44h	-9.8	1101101	6Dh	-13.9	
		0011100	1Ch	-5.8	1000101	45h	-9.9	1101110	6Eh	-14	
		0011101	1Dh	-5.9	1000110	46h	-10	1101111	6Fh	-14.1	
		0011110	1Eh	-6	1000111	47h	-10.1	1110000	70h	-14.2	
		0011111	1Fh	-6.1	1001000	48h	-10.2	1110001	71h	-14.3	
		0100000	20h	-6.2	1001001	49h	-10.3	1110010	72h	-14.4	
		0100001	21h	-6.3	1001010	4Ah	-10.4	1110011	73h	-14.5	
		0100010	22h	-6.4	1001011	4Bh	-10.5	1110100	74h	-14.6	
		0100011	23h	-6.5	1001100	4Ch	-10.6	1110101	75h	-14.7	
		0100100	24h	-6.6	1001101	4Dh	-10.7	1110110	76h	-14.8	
		0100101	25h	-6.7	1001110	4Eh	-10.8	1110111	77h	-14.9	
		0100110	26h	-6.8	1001111	4Fh	-10.9	1111000	78h	-15	
		0100111	27h	-6.9	1010000	50h	-11	other		-15	
		0101000	28h	-7	1010001	51h	-7.1				

Notes:

1. VSP_0/VSN_0 voltage output is ± 15 V fixed value.
2. When switching Mode0 or Mode1, the voltage output is:
 Mode0: VSP_0(+15) / VSN_0 (-15) / VSPL_0 (+3~+15)
 Mode1: VSP_1(+3 ~ +15) / VSN_1(-3 ~ -15) / VSPL_1(+3 ~ +15)

	Mode0	Mode1
VSP	VSP_0(+15)	VSP_1(+3~+15)
VSN	VSN_0(-15)	VSN_1(-3~-15)
VSPL	VSPL_0(+3~+15)	VSPL_1(+3~+15)

3. If gate voltage is set to ± 15 v, ± 10 v, IC will auto correct source voltage as follows

I. VGP- VSP_0 / VSPL_0 / VSP_1 / VSPL_1 ≥ 2 v

II. VGN- VSN_0 / VSN_1 ≥ -2 v

For example:

	symbol	Voltage setting	Real Voltage
Voltage	VGP	10v	+10v
	VGN	10v	-10v
	VSP_0	+15v	+8v
	VSN_0	-15v	-8v
	VSP_1	+5v	+5v
	VSN_1	-5v	-5v
	VSPL	+15v	+8v
	VCOMH	+15v+(-2v)	+8v +(-2v)
	VCOML	-15v+(-2v)	-8v +(-2v)
	VCOMDC	-2v	-2v

4. Voltage setting limit: VSP_0 $\geq$ VSPL_0 , VSP_1 $\geq$ VSPL_1

Restriction

3.1.3 R02H(POF): Power OFF Command

R02H	Bit										
Inst/Para	R/W	D/CX	D7	D6	D5	D4	D3	D2	D1	D0	Code
POF	W	0	0	0	0	0	0	0	1	0	02H
1 st Parameter	W	1	-	-	-	-	-	-	-	EDSE	00

NOTE: “-” Don't care, can be set to VDD or GND level

Description	-The command defines as : ● After power off command, driver will power off base on power off sequence. ● After power off command, BUSY_N signal will drop from high to low. When finish the power off sequence, BUSY_N signal will rise from low to high. ● Power off command will turn off charge pump, T-con, source driver, gate driver, VCOM, temperature sensor, but register and SRAM data will keep until VDD off. ● SD output and VCOM will base on previous condition. It may have two conditions: 0v or floating. 1st parameter <table><tr><th>Bit</th><th>Name</th><th>Description</th></tr><tr><td>0</td><td>EDSE</td><td>EPD Discharge Trigger 0 : Disable EPD discharge (default) 1 : Enable EPD discharge</td></tr></table>			Bit	Name	Description	0	EDSE	EPD Discharge Trigger 0 : Disable EPD discharge (default) 1 : Enable EPD discharge
Bit	Name	Description							
0	EDSE	EPD Discharge Trigger 0 : Disable EPD discharge (default) 1 : Enable EPD discharge							
Restriction	This command only active when BUSY_N = “1”.								

3.1.4 R03H(PFS): Power off Sequence Setting Register

R03H	Bit										
Inst/Para	R/W	D/CX	D7	D6	D5	D4	D3	D2	D1	D0	Code
PFS	W	0	0	0	0	0	0	0	1	1	03H
1 st Parameter	W	1	-		T_VDPG_OFF [1:0]		-		T_VDS_OFF [1:0]		00h
2 nd Parameter	W	1	VGP_LEN[3:0]				VGP_EXT[3:0]				54h
3 rd Parameter	W	1	XON_DLY[3:0]				XON_LEN[3:0]				44h

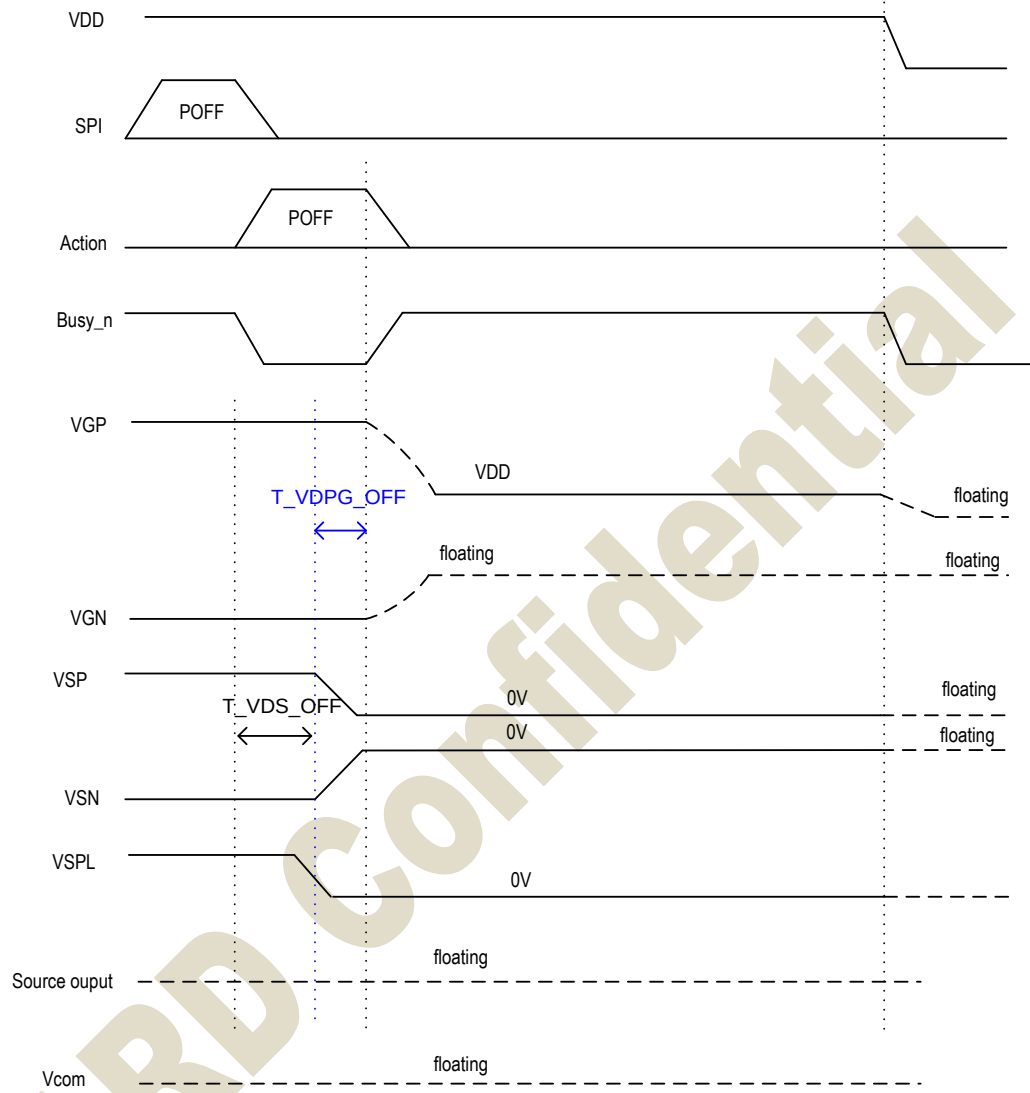
NOTE: "-" Don't care, can be set to VDD or GND level

Description	-The command defines as :		
	1 st Parameter:		
	Bit	Name	Description
	1-0	T_VDS_OFF	Power off sequence of VSP /VSN 00: 20 ms (default) 01: 40 ms 10: 60 ms 11: 80 ms
	5-4	T_VDPG_OFF	Power off sequence of VGP and VGN 00: 20 ms (default) 01: 40 ms 10: 60 ms 11: 80 ms
	2 nd Parameter		
	Bit	Name	Description
	1-0	VGP_EXT	VGP extension time 0000: 0 ms 0001: 500 ms 0010: 1000 ms 0011: 1500 ms 0100: 2000 ms (default) 0101: 2500 ms 0110: 3000 ms 0111: 3500 ms 1000: 4000 ms 1001: 4500 ms 1010: 5000 ms 1011: 5500 ms 1100: 6000 ms 1101: 6500 ms
	7-4	VGP_LEN	When power off, the length of time VGP stay 10V 0000: 0 ms 0001: 500 ms 0010: 1000 ms 0011: 1500 ms 0100: 2000 ms 0101: 2500 ms (default) 0110: 3000 ms 0111: 3500 ms 1000: 4000 ms 1001: 4500 ms 1010: 5000 ms 1011: 5500 ms 1100: 6000 ms 1101: 6500 ms

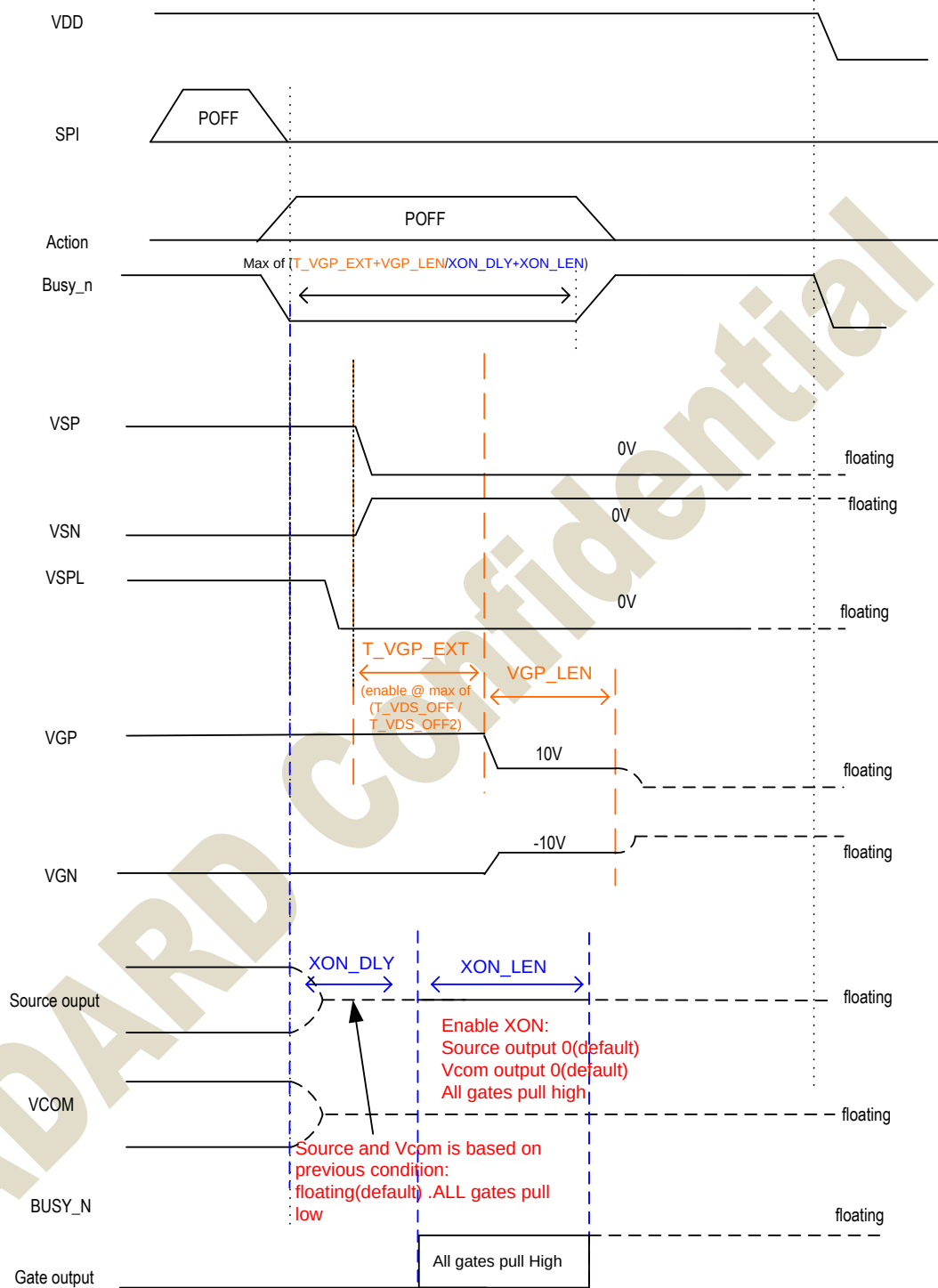
3rd Parameter:

Bit	Name	Description
3-0	XON_LEN	XON enable time 0000: 0 ms 0001: 500 ms 0010: 1000 ms 0011: 1500 ms 0100: 2000 ms (default) 0101: 2500 ms 0110: 3000 ms 0111: 3500 ms 1000: 4000 ms 1001: 4500 ms 1010: 5000 ms 1011: 5500 ms 1100: 6000 ms
7-4	XON_DLY	XON delay time 0000: 0 ms 0001: 500 ms 0010: 1000 ms 0011: 1500 ms 0100: 2000 ms (default) 0101: 2500 ms 0110: 3000 ms 0111: 3500 ms 1000: 4000 ms 1001: 4500 ms 1010: 5000 ms 1011: 5500 ms 1100: 6000 ms

Power off Sequence(R02h=00)



Power off Sequence(R02h=01)



Restriction

3.1.5 R04H (PON): Power ON Command

R04H	Bit										
Inst/Para	R/W	D/CX	D7	D6	D5	D4	D3	D2	D1	D0	Code
PON	W	0	0	0	0	0	0	1	0	0	04H

NOTE: “-” Don't care, can be set to VDD or GND level

Description	-The command defines as : - After power on command, driver will power on base on power on sequence. - After power on command, BUSY_N signal will drop from high to low. When finishing the power on sequence, BUSY_N signal will rise from low to high.
Restriction	This command only active when BUSY_N = “1”.

3.1.6 R06H(BTST): Booster Soft Start Command

R06H	Bit										
Inst/Para	R/W	D/CX	D7	D6	D5	D4	D3	D2	D1	D0	Code
BTST	W	0	0	0	0	0	0	1	1	0	06H
1 st Parameter	W	1	-	-	-	-	PHB_SFT [1:0]		PHA_SFT [1:0]		17h
2 nd Parameter	W	1	-	-	PHA_ON [5:0]						02h
3 rd Parameter	W	1	-	-	PHA_OFF [5:0]						07h
4 th Parameter	W	1	-	-	PHB_ON [5:0]						02h
5 th Parameter	W	1	-	-	PHB_OFF [5:0]						07h
6 th Parameter	W	1	-	-	PHC_ON [5:0]						02h
7 th Parameter	W	1	-	-	PHC_OFF [5:0]						07h

NOTE: "-" Don't care, can be set to VDD or GND level

-The command define as follows:

1st Parameter:

Bit	Name	Description
1-0	PHA_SFT	Soft start period of phase A: 00: 10mS (default) 01: 20mS 10: 30mS 11: 40mS
3-2	PHB_SFT	Soft start period of phase B: 00: 10mS (default) 01: 20mS 10: 30mS 11: 40mS

Description

	Bit[5:0]	Description	Bit[5:0]	Description	Bit[5:0]	Description
	000000	strength1	010110	strength23	101100	strength45
	000001	strength2	010111	strength24	101101	strength46
	000010	strength3	011000	strength25	101110	strength47
	000011	strength4	011001	strength26	101111	strength48
	000100	strength5	011010	strength27	110000	strength49
	000101	strength6	011011	strength28	110001	strength50
	000110	strength7	011100	strength29	110010	strength51
	000111	strength8	011101	strength30	110011	strength52
	001000	strength9	011110	strength31	110100	strength53
	001001	strength10	011111	strength32	110101	strength54
	001010	strength11	100000	strength33	110110	strength55
	001011	strength12	100001	strength34	110111	strength56
	001100	strength13	100010	strength35	111000	strength57
	001101	strength14	100011	strength36	111001	strength58
	001110	strength15	100100	strength37	111010	strength59
	001111	strength16	100101	strength38	111011	strength60
	010000	strength17	100110	strength39	111100	strength61
	010001	strength18	100111	strength40	111101	strength62
	010010	strength19	101000	strength41	111110	strength63
	010011	strength20	101001	strength42	111111	strength64
	010100	strength21	101010	strength43		
	010101	strength22	101011	strength44		

Driving
strength of
PHA_ON &
PHB_ON &
PHC_ON

Description		Bit[5:0]	Description	Bit[5:0]	Description	Bit[5:0]	Description
Minimum OFF time setting of PHA_OFF & PHB_OFF & PHC_OFF		000000	Period1	010110	Period23	101100	Period45
		000001	Period2	010111	Period24	101101	Period46
		000010	Period3	011000	Period25	101110	Period47
		000011	Period4	011001	Period26	101111	Period48
		000100	Period5	011010	Period27	110000	Period49
		000101	Period6	011011	Period28	110001	Period50
		000110	Period7	011100	Period29	110010	Period51
		000111	Period8	011101	Period30	110011	Period52
		001000	Period9	011110	Period31	110100	Period53
		001001	Period10	011111	Period32	110101	Period54
		001010	Period11	100000	Period33	110110	Period55
		001011	Period12	100001	Period34	110111	Period56
		001100	Period13	100010	Period35	111000	Period57
		001101	Period14	100011	Period36	111001	Period58
		001110	Period15	100100	Period37	111010	Period59
		001111	Period16	100101	Period38	111011	Period60
		010000	Period17	100110	Period39	111100	Period61
		010001	Period18	100111	Period40	111101	Period62
		010010	Period19	101000	Period41	111110	Period63
		010011	Period20	101001	Period42	111111	Period64
		010100	Period21	101010	Period43		
		010101	Period22	101011	Period44		
Restriction							

3.1.7 R07H(DSLP): Deep Sleep Command

R07H	Bit										
Inst/Para	R/W	D/CX	D7	D6	D5	D4	D3	D2	D1	D0	Code
DSL P	W	0	0	0	0	0	0	1	1	1	07H
1 st Parameter	W	1	1	0	1	0	0	1	0	1	A5h

NOTE: "-" Don't care, can be set to VDD or GND level

Description	The command define as follows: After this command is transmitted, the chip would enter the deep-sleep mode to save power. The deep sleep mode would return to standby by hardware reset. The only one parameter is a check code, the command would be excited if check code = 0xA5.
Restriction	This command only active when BUSY_N = "1".

3.1.8 R10H(DTM1): Data Start Transmission 1 Register

R10H	Bit										
Inst/Para	R/W	D/CX	D7	D6	D5	D4	D3	D2	D1	D0	Code
DTM1	W	0	0	0	0	1	0	0	0	0	10H
1 st Parameter	W	1	Pixel1		Pixel2		Pixel3		Pixel4		00h
2 nd Parameter	W	1	⋮		⋮		⋮		⋮		00h
...	W	1	Pixel(n-3)		Pixel(n-2)		Pixel(n-1)		Pixel(n)		00h
M th Parameter	W	1	Pixel1		Pixel2		Pixel3		Pixel4		00h

NOTE: "-" Don't care, can be set to VDD or GND level

Description	The command define as follows: The register is indicates that user start to transmit data, then write to SRAM. While data transmission complete, user must send command 12H. Then chip will start to send data/VCOM for panel. Pixel [1~n][1:0]: 2-bit/pixel <table> <tr> <th>Image Data</th><th colspan="2">DDX=1(default)</th><th colspan="2">DDX=0</th></tr> <tr> <th>Pixel[1:0]</th><th>Gray level select</th><th>IP output LUT select</th><th>Gray level select</th><th>IP output LUT select</th></tr> <tr> <td>00b</td><td>Gray0</td><td>ogray00</td><td>Gray3</td><td>ogray03</td></tr> <tr> <td>01b</td><td>Gray1</td><td>ogray01</td><td>Gray2</td><td>ogray02</td></tr> <tr> <td>10b</td><td>Gray2</td><td>ogray02</td><td>Gray1</td><td>ogray01</td></tr> <tr> <td>11b</td><td>Gray3</td><td>ogray03</td><td>Gray0</td><td>ogray00</td></tr> </table>				Image Data	DDX=1(default)		DDX=0		Pixel[1:0]	Gray level select	IP output LUT select	Gray level select	IP output LUT select	00b	Gray0	ogray00	Gray3	ogray03	01b	Gray1	ogray01	Gray2	ogray02	10b	Gray2	ogray02	Gray1	ogray01	11b	Gray3	ogray03	Gray0	ogray00
Image Data	DDX=1(default)		DDX=0																															
Pixel[1:0]	Gray level select	IP output LUT select	Gray level select	IP output LUT select																														
00b	Gray0	ogray00	Gray3	ogray03																														
01b	Gray1	ogray01	Gray2	ogray02																														
10b	Gray2	ogray02	Gray1	ogray01																														
11b	Gray3	ogray03	Gray0	ogray00																														
Restriction																																		

3.1.9 R12H(DRF): Display Refresh Command

R12H	Bit										
Inst/Para	R/W	D/CX	D7	D6	D5	D4	D3	D2	D1	D0	Code
DRF	W	0	0	0	0	1	0	0	1	0	12H
1 st Parameter	W	1	-	-	-	-	-	-	-	AC/DC VCOM	00h

NOTE: "-" Don't care, can be set to VDD or GND level

Description	-The command defines as : While users send this command, driver will refresh display (data/VCOM) base on SRAM data and LUT. AC/DC VCOM: 0: AC VCOM, VCOM will follow LUTC when updating image. (default) 1: DC VCOM, VCOM will always be VCOMDC when updating image After display refresh command, BUSY_N signal will become "0"
Restriction	This command only actives when BUSY_N = "1"

3.1.10 R17H(AUTO): Auto Sequence

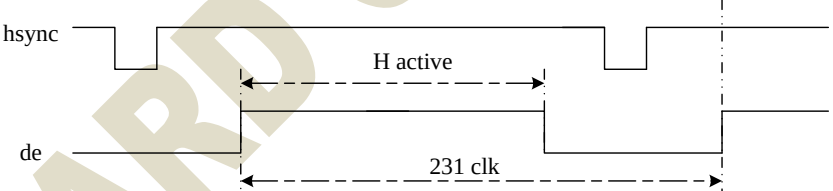
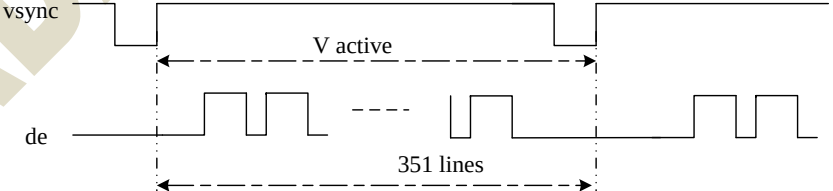
R17H	Bit										
Inst/Para	R/W	D/CX	D7	D6	D5	D4	D3	D2	D1	D0	Code
Auto Sequence	W	0	0	0	0	1	0	1	1	1	17H
1 st Parameter	W	1	Code[7]	Code[6]	Code[5]	Code[4]	Code[3]	Code[2]	Code[1]	Code[0]	A5h

Description	The command can enable the internal sequence to execute several commands continuously. The successive execution can minimize idle time to avoid unnecessary power consumption and reduce the complexity of host's control procedure. The sequence contains several operations, including PON, DRF, POF, DSLP. AUTO (0x17) + Code(0xA5) = (PON→DRF→POF) AUTO (0x17) + Code(0xA7) = (PON→DRF→POF→DSLP)
Restriction	This command only actives when BUSY_N = "1".

3.1.11 R30H(PLL): PLL Control Register

R30H	Bit										
Inst/Para	R/W	D/CX	D7	D6	D5	D4	D3	D2	D1	D0	Code
PLL	W	0	0	0	1	1	0	0	0	0	30H
1 st Parameter	W	1	-	-	-	-	Dyna	FR[2:0]			02h

NOTE: “-” Don't care, can be set to VDD or GND level

Description	-The command defines as: The command controls the PLL clock frequency. The PLL structure must support the following frame rates: <table border="1"> <thead> <tr> <th>bit3</th><th>Dynamic frame rate</th></tr> </thead> <tbody> <tr> <td>0</td><td>Disable(default)</td></tr> <tr> <td>1</td><td>Enable</td></tr> </tbody> </table> <table border="1"> <thead> <tr> <th>FR[2:0]</th><th>Frame rate</th></tr> </thead> <tbody> <tr> <td>000</td><td>12.5 Hz</td></tr> <tr> <td>001</td><td>25 Hz</td></tr> <tr> <td>010</td><td>50 Hz(default)</td></tr> <tr> <td>011</td><td>65 Hz</td></tr> <tr> <td>100</td><td>75 Hz</td></tr> <tr> <td>101</td><td>85 Hz</td></tr> <tr> <td>110</td><td>100 Hz</td></tr> <tr> <td>111</td><td>120 Hz</td></tr> </tbody> </table>	bit3	Dynamic frame rate	0	Disable(default)	1	Enable	FR[2:0]	Frame rate	000	12.5 Hz	001	25 Hz	010	50 Hz(default)	011	65 Hz	100	75 Hz	101	85 Hz	110	100 Hz	111	120 Hz
bit3	Dynamic frame rate																								
0	Disable(default)																								
1	Enable																								
FR[2:0]	Frame rate																								
000	12.5 Hz																								
001	25 Hz																								
010	50 Hz(default)																								
011	65 Hz																								
100	75 Hz																								
101	85 Hz																								
110	100 Hz																								
111	120 Hz																								
remark	-Horizontal  -Vertical 																								
Restriction																									

3.1.12 R40H(TSC): Temperature Sensor Command

R40H	Bit										
Inst/Para	R/W	D/CX	D7	D6	D5	D4	D3	D2	D1	D0	Code
TSC	W	0	0	1	0	0	0	0	0	0	40H
1 st Parameter	R	1	D10/TS[9]	D9/TS[8]	D8/TS[7]	D7/TS[6]	D6/TS[5]	D5/TS[4]	D4/TS[3]	D3/TS[2]	-
2nd Parameter	R	1	D2/TS[1]	D1/TS[0]	D0	-	-	-	-	-	-

NOTE: "-" Don't care, can be set to VDD or GND level

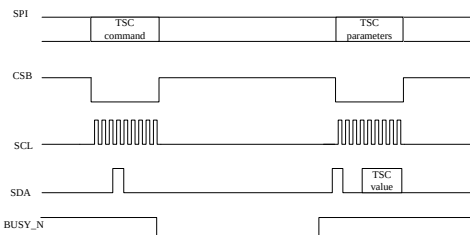
Description

-The command define as follows:

This command indicates the temperature value.

If R41H(TSE) bit7 set to 0, this command reads internal temperature sensor value.

If R41H(TSE) bit7 set to 1, this command reads external (LM75) temperature sensor value



TS[9:2]/D[10:3]	T (°C)	TS[9:2]/D[10:3]	T (°C)	TS[9:2]/D[10:3]	T (°C)
11100111	-25	00000000	0	00011001	25
11101000	-24	00000001	1	00011010	26
11101001	-23	00000010	2	00011011	27
11101010	-22	00000011	3	00011100	28
11101011	-21	00000100	4	00011101	29
11101100	-20	00000101	5	00011110	30
11101101	-19	00000110	6	00011111	31
11101110	-18	00000111	7	00100000	32
11101111	-17	00001000	8	00100001	33
11110000	-16	00001001	9	00100010	34
11110001	-15	00001010	10	00100011	35
11110010	-14	00001011	11	00100100	36
11110011	-13	00001100	12	00100101	37
11110100	-12	00001101	13	00100110	38
11110101	-11	00001110	14	00100111	39
11110110	-10	00001111	15	00101000	40
11110111	-9	00010000	16	00101001	41
11111000	-8	00010001	17	00101010	42
11111001	-7	00010010	18	00101011	43
11111010	-6	00010011	19	00101100	44
11111011	-5	00010100	20	00101101	45
11111100	-4	00010101	21	00101110	46
11111101	-3	00010110	22	00101111	47
11111110	-2	00010111	23	00110000	48
11111111	-1	00011000	24	00110001	49

TS[1:0]	T (°C)
00	+0
01	+0.25
10	+0.5
11	+0.75

Restriction

This command only actives when BUSY_N = "1".

3.1.13 R41H(TSE): Temperature Sensor Calibration Register

R41H	Bit										
Inst/Para	R/W	D/CX	D7	D6	D5	D4	D3	D2	D1	D0	Code
TSE	W	0	0	1	0	0	0	0	0	1	41H
1 st Parameter	W	1	TSE	-	TO[5]	TO[4]	TO[3]	TO[2]	TO[1]	TO[0]	00h

NOTE: "-" Don't care, can be set to VDD or GND level

Description	-The command defines as: This command indicates the driver IC temperature sensor enable and calibration function.	
	Reserve one temperature offset TO[3:0] for calibration 1. TO[3]: mean '+' or '-', while 0 is '+' ; 1 is '-' 2. TO[2:0]: mean temperature offset value	
	Bit	Name
	3-0	TO[3:0]
Description	Temperature level: 0000: +0°C (default) 0001: +0.5°C 0010: +1°C 0011: +1.5°C 0100: +2°C 0101: +2.5°C 0110: +3°C 0111: +3.5°C 1000: -4°C 1001: -3.5°C 1010: -3°C 1011: -2.5°C 1100: -2°C 1101: -1.5°C 1110: -1°C 1111: -0.5°C	
	5-4	TO[5:4]
	0: +0.0°C (default) 1: +0.25°C	
Description	Internal temperature sensor enable	
	0: Internal temperature sensor enable.(default) 1: Internal temperature sensor disable, using external temperature sensor.	
Restriction	This command only actives after R04H(PON)	

3.1.14 R50H(CDI): VCOM and DATA Interval Setting Register

R50H	Bit										
Inst/Para	R/W	D/CX	D7	D6	D5	D4	D3	D2	D1	D0	Code
CDI	W	0	0	1	0	1	0	0	0	0	50H
1 st Parameter	W	1	VBD[2]	VBD[1]	VBD [0]	DDX	CDI[3]	CDI[2]	CDI[1]	CDI[0]	97h

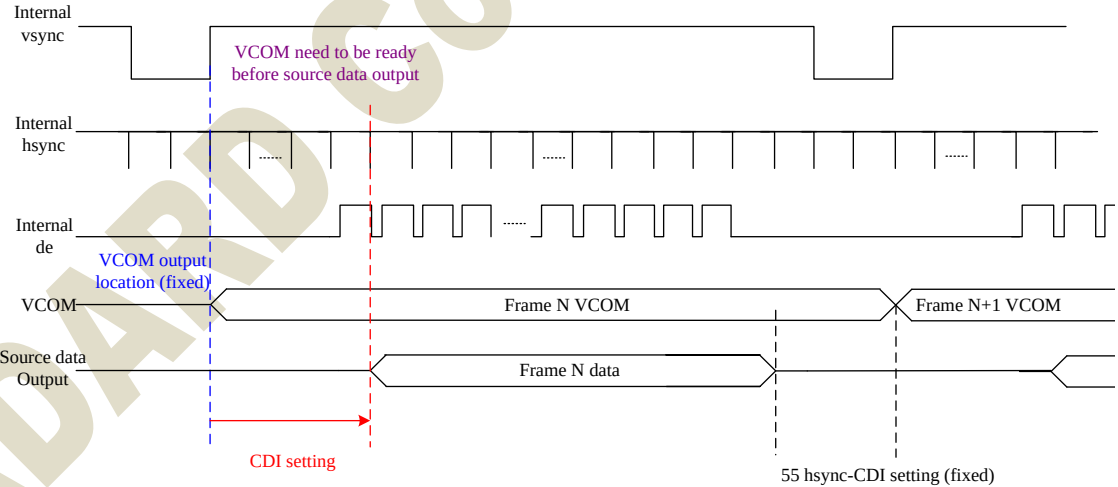
NOTE: "-" Don't care, can be set to VDD or GND level

Description

-The command defines as:
1st Parameter:

CDI[1:0]: This command indicates the interval of VCOM and data output. When setting the vertical back porch, **the total blanking will be keep (55hsync).**

Bit	Name	Description
3-0	CDI[3:0]	Vcom and data interval 0000: 17 hsync 0001:16 hsync 0010:15 hsync 0011:14 hsync 0100:13 hsync 0101:12 hsync 0110:11 hsync 0111:10 hsync(default) 1000:9 hsync 1001:8 hsync 1010:7 hsync 1011:6 hsync 1100:5 hsync 1101:4 hsync 1110:3 hsync 1111:2 hsync



VBD[2:0]: Border data selection. (from LUT output by IP port border_w[1:0])

This register will make boarder pin output being mapped to a certain gray scale.

Bit 4	Bit7-5	Description	IP setting for Border LUT select
DDX	VBD[2:0]	Gray level	
0	000	Floating	N/A
	001	Gray3	border_buf=011
	010	Gray2	border_buf=010
	011	Gray1	border_buf=001
	100	Gray0	border_buf=000
1 (default)	000	Gray0	border_buf=000
	001	Gray1	border_buf=001
	010	Gray2	border_buf=010
	011	Gray3	border_buf=011
	100	Floating	N/A

Border output voltage level: The level selection is based on mapping LUT data.

Ex: Gray 1 waveform is mapping to 15V, without VCOM offset, the real output on Boarder pin shall be 15V.

Boarder output will follow FOPT definition being defined in R00h.

Restriction

3.1.15 R51H(LPD): Lower Power Detection Register

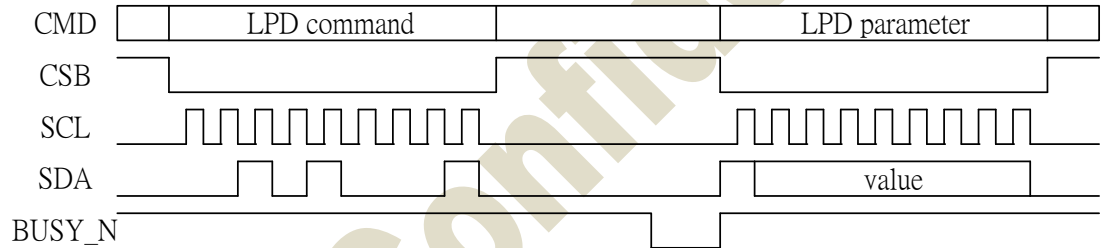
R51H	Bit										
Inst/Para	R/W	D/CX	D7	D6	D5	D4	D3	D2	D1	D0	Code
LPD	W	0	0	1	0	1	0	0	0	1	51H
1 st Parameter	R	1	-	-	-	-	-	-	-	LPD	--

NOTE: "-" Don't care, can be set to VDD or GND level

Description -The command defines as:
This command indicates the input power condition. Host can read this data to understand the battery's condition.
When LPD="1", system input power is normal.
When LPD="0", system input power is lower ($V_{DD} < 2.5V$, which could be select in RE4H (LVSEL)).

1st Parameter:

Bit	Name	Description
0	LPD	0: Low power input 1: Normal status



Restriction - This command only activates when BUSY_N = "1".

3.1.16 R61H(TRES): Resolution Setting

R61H	Bit										
Inst/Para	R/W	D/CX	D7	D6	D5	D4	D3	D2	D1	D0	Code
TRES	W	0	0	1	1	0	0	0	0	1	61H
1 st Parameter	W	1	-	-	-	-	-	-	HRES(9)	HRES(8)	00h
2 nd Parameter	W	1	HRES(7)	HRES(6)	HRES(5)	HRES(4)	HRES(3)	HRES(2)	-	-	00h
3 rd Parameter	W	1	-	-	-	-	-	-	-	-	00h
4 th Parameter	W	1	VRES(7)	VRES(6)	VRES(5)	VRES(4)	VRES(3)	VRES(2)	VRES(1)	VRES(0)	00h

NOTE: "-" Don't care, can be set to VDD or GND level

Description	-The command define as follows: When using register: Horizontal display resolution = HRES Vertical display resolution = VRES Channel disable calculation: GD : First G active = G0; LAST active GD= first active +VRES[8:0] -1 SD : First active channel: =S0 ; LAST active SD= first active +HRES[7:3]*8-1 EX :176X296 GD: First G active = G0 LAST active GD= 0+296-1= 295; (G295) SD : First active channel: =S0 LAST active SD=0+44*4-1=175; (S175) R61H = 00h,98h, 01h, 28h (解析度直接轉 16 進制即為設定值) Note: Only supports source 176.ch for source 160ch. above
Restriction	

3.1.17 R65H(GSST): Gate/Source Start Setting Register

R65H	Bit										
Inst/Para	R/W	D/CX	D7	D6	D5	D4	D3	D2	D1	D0	Code
GSST	W	0	0	1	1	0	0	1	0	1	65H
1 st Parameter	W	1	-	-	-	-	-	-	S_start[9]	S_start[8]	00h
2 nd Parameter	W	1	S_start[7]	S_start[6]	S_start[5]	S_start[4]	S_start[3]	S_start[2]	-	-	00h
3 rd Parameter	W	1	-	-	-	-	-	-	G_start[9]	G_start[8]	00h
4 th Parameter	W	1	G_start[7]	G_start[6]	G_start[6]	G_start[4]	G_start[3]	G_start[2]	G_start[1]	G_start[0]	00h

NOTE: "-" Don't care, can be set to VDD or GND level

Description	-The command define as follows: 1.S_Start [7:0] describe which source output line is the first date line 2.G_Start[8:0] describe which gate line is the first scan line
Restriction	

3.1.18 R70H(REV): REVISION Register

R70H	Bit										
Inst/Para	R/W	D/CX	D7	D6	D5	D4	D3	D2	D1	D0	Code
REV	W	0	0	1	1	1	0	0	0	0	70H
1 st Parameter	R	1	0	0	0	0	0	0	1	1	03h
2 nd Parameter	R	1	0	0	0	0	0	0	1	0	02h
3 rd Parameter	R	1	0	0	0	0	0	0	0	1	01h

NOTE: "-" Don't care, can be set to VDD or GND level

Description	-The command defines as:	
	The LUT_REV is read from:	
	1 st & 2 nd & 3 rd Parameter :	
	Bit	Description
	3-0	CHIP_REV
Restriction		

3.1.19 R80H(AMV): Auto Measure VCOM Register

R80H	Bit										
Inst/Para	R/W	D/CX	D7	D6	D5	D4	D3	D2	D1	D0	Code
AMV	W	0	1	0	0	0	0	0	0	0	80H
1 st Parameter	W	1	-	-	AMVT[1]	AMVT[0]	XON	AMVS	AMV	AMVE	10h

NOTE: "-" Don't care, can be set to VDD or GND level

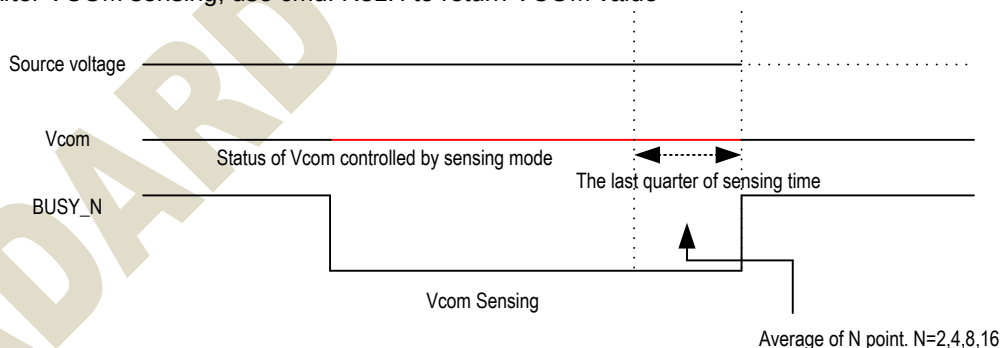
Description

-The command defines as:
This command indicates the IC status. Host can read this data to understand the IC status.

1st Parameter:

Bit	Name	Description
0	AMVE	AMVE: Auto Measure Vcom Setting 0: Auto measure VCOM disable (default) 1: Auto measure VCOM enable
1	AMV	AMV: Analog signal 0: Get Vcom value from R81h(default) 1: Get Vcom value in analog signal
2	AMVS	AMVS: setting for Source output of AMV 0: Source output 0V during Auto Measure VCOM period. (default) 1: Source output VSPL_0 during Auto Measure VCOM period.
3	XON	XON: setting for all Gate ON of AMV 0: Gate normally scan during Auto Measure VCOM period. (default) 1: All Gate ON during Auto Measure VCOM period.
5-4	AMVT[1:0]	The sensing time of VCOM detection 00: 5s (default) 01: 10s 10: 15s 11: 20s

After VCOM sensing, use cmd. R81H to return VCOM value



Note:

- 1.VCOM設定至最小0V(R82h=80h)
- 2.外部穩壓電容floating

Restriction

This command only actives when BUSY_N = "1".

3.1.20 R81H(VV): VCOM Value register

R81H	Bit										
Inst/Para	R/W	D/CX	D7	D6	D5	D4	D3	D2	D1	D0	Code
VV	W	0	1	0	0	0	0	0	0	1	81H
1 st Parameter	R	1	-	-	VV[5]	VV[4]	VV[3]	VV[2]	VV[1]	VV[0]	--

NOTE: "-" Don't care, can be set to VDD or GND level

Description -The command defines as:
This command could get the VCOM value

1st Parameter:

Bit	Name	Description					
6-0	VV[6:0]	VCOM value					
		VCOM[6:0]	Voltage(V)	VCOM[6:0]	Voltage(V)	VCOM[6:0]	Voltage(V)
		0000000	00h 0	0011100	1Ch -1.4	0111000	38h -2.8
		0000001	01h -0.05	0011101	1Dh -1.45	0111001	39h -2.85
		0000010	02h -0.1	0011110	1Eh -1.5	0111010	3Ah -2.9
		0000011	03h -0.15	0011111	1Fh -1.55	0111011	3Bh -2.95
		0000100	04h -0.2	0100000	20h -1.6	0111100	3Ch -3
		0000101	05h -0.25	0100001	21h -1.65	0111101	3Dh -3.05
		0000110	06h -0.3	0100010	22h -1.7	0111110	3Eh -3.1
		0000111	07h -0.35	0100011	23h -1.75	0111111	3Fh -3.15
		0001000	08h -0.4	0100100	24h -1.8	1000000	40h -3.2
		0001001	09h -0.45	0100101	25h -1.85	1000001	41h -3.25
		0001010	0Ah -0.5	0100110	26h -1.9	1000010	42h -3.3
		0001011	0Bh -0.55	0100111	27h -1.95	1000011	43h -3.35
		0001100	0Ch -0.6	0101000	28h -2	1000100	44h -3.4
		0001101	0Dh -0.65	0101001	29h -2.05	1000101	45h -3.45
		0001110	0Eh -0.7	0101010	2Ah -2.1	1000110	46h -3.5
		0001111	0Fh -0.75	0101011	2Bh -2.15	1000111	47h -3.55
		0010000	10h -0.8	0101100	2Ch -2.2	1001000	48h -3.6
		0010001	11h -0.85	0101101	2Dh -2.25	1001001	49h -3.65
		0010010	12h -0.9	0101110	2Eh -2.3	1001010	4Ah -3.7
		0010011	13h -0.95	0101111	2Fh -2.35	1001011	4Bh -3.75
		0010100	14h -1	0110000	30h -2.4	1001100	4Ch -3.8
		0010101	15h -1.05	0110001	31h -2.45	1001101	4Dh -3.85
		0010110	16h -1.1	0110010	32h -2.5	1001110	4Eh -3.9
		0010111	17h -1.15	0110011	33h -2.55	1001111	4Fh -3.95
		0011000	18h -1.2	0110100	34h -2.6	1010000	50h -4
		0011001	19h -1.25	0110101	35h -2.65	other	-4
		0011010	1Ah -1.3	0110110	36h -2.7		
		0011011	1Bh -1.35	0110111	37h -2.75		

Restriction

3.1.21 R82H(VDCS): VCOM_DC Setting Register

R82H	Bit										
Inst/Para	R/W	D/CX	D7	D6	D5	D4	D3	D2	D1	D0	Code
VDCS	W	0	1	0	0	0	0	0	1	0	82H
1 st Parameter	W	1	MTP_VCM	VDCS[6]	VDCS[5]	VDCS [4]	VDCS [3]	VDCS [2]	VDCS [1]	VDCS [0]	00h

NOTE: "-" Don't care, can be set to VDD or GND level

Description

-The command defines as:
 This command set the VCOM DC value. Driver will base on this value for VCM_DC.
 1st Parameter:
 1st Parameter:

Bit	Name	Description					
6-0	VDCS[6:0]	VCOM value					
		VCOM[6:0]	Voltage(V)	VCOM[6:0]	Voltage(V)	VCOM[6:0]	Voltage(V)
		0000000	00h	0(default)	0011100	1Ch	-1.4
		0000001	01h	-0.05	0011101	1Dh	-1.45
		0000010	02h	-0.1	0011110	1Eh	-1.5
		0000011	03h	-0.15	0011111	1Fh	-1.55
		0000100	04h	-0.2	0100000	20h	-1.6
		0000101	05h	-0.25	0100001	21h	-1.65
		0000110	06h	-0.3	0100010	22h	-1.7
		0000111	07h	-0.35	0100011	23h	-1.75
		0001000	08h	-0.4	0100100	24h	-1.8
		0001001	09h	-0.45	0100101	25h	-1.85
		0001010	0Ah	-0.5	0100110	26h	-1.9
		0001011	0Bh	-0.55	0100111	27h	-1.95
		0001100	0Ch	-0.6	0101000	28h	-2
		0001101	0Dh	-0.65	0101001	29h	-2.05
		0001110	0Eh	-0.7	0101010	2Ah	-2.1
		0001111	0Fh	-0.75	0101011	2Bh	-2.15
		0010000	10h	-0.8	0101100	2Ch	-2.2
		0010001	11h	-0.85	0101101	2Dh	-2.25
		0010010	12h	-0.9	0101110	2Eh	-2.3
		0010011	13h	-0.95	0101111	2Fh	-2.35
		0010100	14h	-1	0110000	30h	-2.4
		0010101	15h	-1.05	0110001	31h	-2.45
		0010110	16h	-1.1	0110010	32h	-2.5
		0010111	17h	-1.15	0110011	33h	-2.55
		0011000	18h	-1.2	0110100	34h	-2.6
		0011001	19h	-1.25	0110101	35h	-2.65
		0011010	1Ah	-1.3	0110110	36h	-2.7
		0011011	1Bh	-1.35	0110111	37h	-2.75
				other		-4	

	7	MTP_VCM	Follow MTP VCOM value in MTP mode 0: From the setting of MTP (default) 1: From the setting of register
Restriction			

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3.1.22 R83H(PTL): Partial Window Register

R83H	Bit										
Inst/Para	R/W	D/CX	D7	D6	D5	D4	D3	D2	D1	D0	Code
PTL	W	0	1	0	0	0	0	0	1	1	83H
1 st Parameter	W	1	-	-	-	-	-	-	HRST[9]	HRST[8]	00h
2 nd Parameter	W	1	HRST[7]	HRST[6]	HRST[5]	HRST[4]	HRST[3]	HRST[2]	-	-	00h
3 rd Parameter	W	1	-	-	-	-	-	-	HRED[9]	HRED[8]	00h
4 th Parameter	W	1	HRED[7]	HRED[6]	HRED[5]	HRED[4]	HRED[3]	HRED[2]	-	-	00h
5 th Parameter	W	1	-	-	-	-	-	-	VRST[9]	VRST[8]	00h
6 th Parameter	W	1	VRST[7]	VRST[6]	VRST[5]	VRST[4]	VRST[3]	VRST[2]	VRST[1]	VRST[0]	00h
7 th Parameter	W	1	-	-	-	-	-	-	VRED[9]	VRED[8]	00h
8 th Parameter	W	1	VRED[7]	VRED[6]	VRED[5]	VRED[4]	VRED[3]	VRED[2]	VRED[1]	VRED[0]	00h
9 th Parameter	W	1	-	-	-	-	-	-	-	PMODE	00h

NOTE: "-" Don't care, can be set to VDD or GND level

Description -This command sets partial window.

Name	Description
HRST[9:2]	Horizontal start address
HRED[9:2]	Horizontal end address. HRED must be greater than HRST.
VRST[9:0]	Vertical start address.
VRED[9:0]	Vertical end address. VRED must be greater than VRST.
PMODE	0: disable partial mode(default) 1: enable partial mode

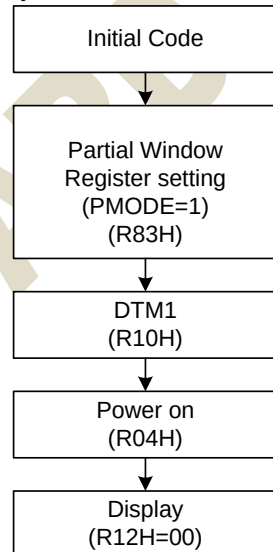
Note:

No matter HRST[1:0], HRST[9:8], HRED[9:8], VRST[9], VRED[9] value being filled, it's always be 00b.

No matter HRED[1:0] value being filled, it's always be 11b.

Gates scan both inside and outside of the partial window.

Partial display flow:



Restriction

3.1.23 R90H(PGM): Program Mode

R90H	Bit										
Inst/Para	R/W	D/CX	D7	D6	D5	D4	D3	D2	D1	D0	Code
PGM	W	0	1	0	1	0	0	0	0	0	A0H

NOTE: “-” Don’t care, can be set to VDD or GND level

Description	-The command define as follows: After this command is issued, the chip would enter the program mode. The mode would return to standby by hardware reset.
Restriction	This command only actives when BUSY_N = “1”.

3.1.24 R91H(APG): Active Program

R91H	Bit										
Inst/Para	R/W	D/CX	D7	D6	D5	D4	D3	D2	D1	D0	Code
APG	W	0	1	0	0	1	0	0	0	1	91H

NOTE: "-" Don't care, can be set to VDD or GND level

Description	-The command define as follows: After this command is transmitted, the programming state machine would be activated.
Restriction	The BUSY flag would change state from 0 to 1 while the programming is completed.

3.1.25 R92H(RMTP): Read MTP Data

R92H	Bit										
Inst/Para	R/W	D/CX	D7	D6	D5	D4	D3	D2	D1	D0	Code
RMTP	W	0	1	0	0	1	0	0	1	0	92H
1 st Parameter	R	1	Dummy								-
2 nd Parameter	R	1	The data of address 0x000 in the MTP								-
3 rd Parameter	R	1	The data of address 0x001 in the MTP								-
4 th Parameter	R	1	:								-
5 th Parameter	R	1	The data of address (n-1) in the MTP								-
6 th ~(m-1) th Parameter	R	1									-
m th Parameter	R	1	The data of address (n) in the MTP								-

NOTE: “-” Don't care, can be set to VDD or GND level

Description	-The command define as follows: The command is used for reading the content of MTP for checking the data of programming. The value of (n) is depending on the amount of programmed data, the max address = 0x17FF.
Restriction	This command only actives when BUSY_N = “1”.

3.1.26 R9FH(RMRB): Read MTP Reserved Bytes

R9FH	Bit										
Inst/Para	R/W	D/CX	D7	D6	D5	D4	D3	D2	D1	D0	Code
RMRB	W	0	1	0	0	1	1	1	1	1	9FH
1 st Parameter	R	1	Dummy								-
2 nd Parameter	R	1	The data of address 0x16F7 in the MTP								00h
3 rd Parameter	R	1	:								00h
:	R	1	:								00h
97 th Parameter	R	1	:								00h
98 th Parameter	R	1	:								00h
101 th Parameter	R	1	The data of address 0x175A in the MTP								00h

NOTE: “-” Don't care, can be set to VDD or GND level

Description	-The command define as follows: The command is used for reading the content of MTP Reserved Byte for checking the data of programming. This command could read these information from MTP directly.
Restriction	This command only actives when BUSY_N = “1”.

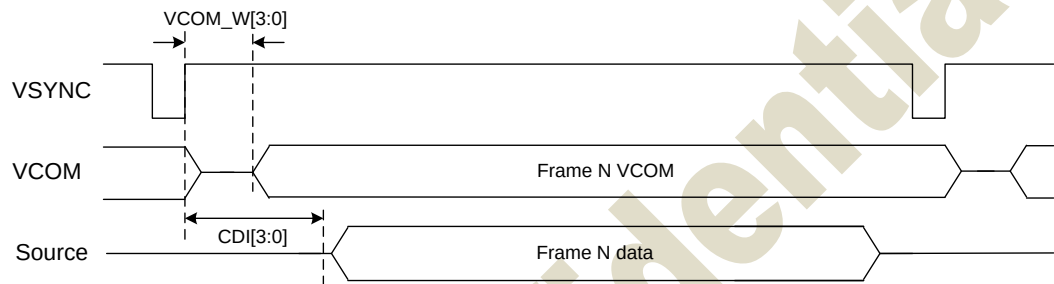
3.1.27 RE3H(PWS): Power Saving Register

RE3H	Bit										
Inst/Para	R/W	D/CX	D7	D6	D5	D4	D3	D2	D1	D0	Code
PWS	W	0	1	1	1	0	0	0	1	1	E3H
1 st Parameter	W	1	VCOM_W[3:0]				SD_W[3:0]				00h

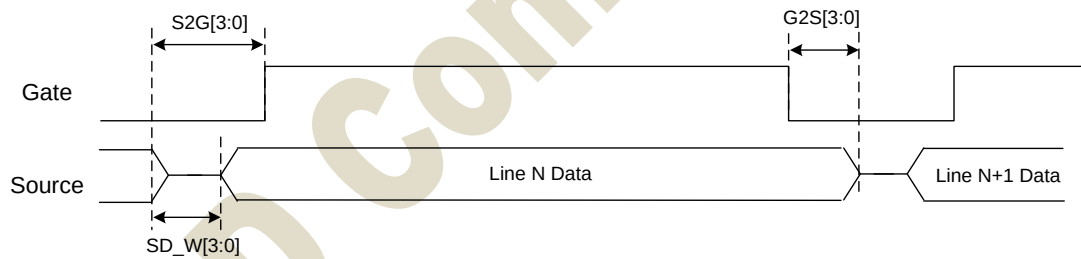
NOTE: "-" Don't care, can be set to VDD or GND level

Description - This command is set for saving power during refreshing period. If the output voltage of VCOM / Source is from negative to positive or from positive to negative, the power saving mechanism will be activated. The active period width is defined by the following two parameters.

VCOM_W: VCOM power saving width (unit = line period)



SD_W: Source power saving width (unit = 660nS)



Restriction

3.1.28 RE4H (LVSEL): LVD Voltage Select Register

RE4H	Bit										
Inst/Para	R/W	D/CX	D7	D6	D5	D4	D3	D2	D1	D0	Code
LVSEL	W	0	1	1	1	0	0	1	0	0	E4H
1 st Parameter	W	1	-	-	-	-	-	-	LVD_SEL[1:0]		03h

NOTE: "-" Don't care, can be set to VDD or GND level

Description	LVD_SEL[1:0]: Low Power Voltage Selection	
	LVD_SEL[1:0]	LVD value
	00	< 2.2 V
	01	< 2.3 V
	10	< 2.4 V
	11	< 2.5 V (default)
Restriction		

3.2 PWD

PWD 為 IC 內部相關功能控制的 register；使用 PWD 前皆需先下 0x4Dh=78h 後方可使用，相關內容可參考下表。

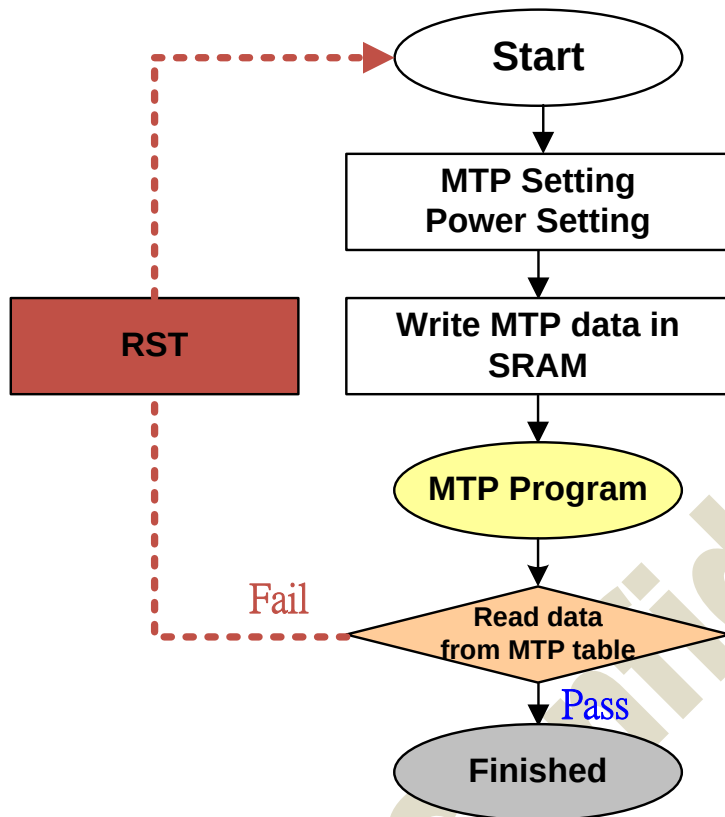
W/R	Address (Hex)	Data (Hex)	Description
W	4D	78	Enter PWD

3.3 IC Vender ID Read

W/R	Address (Hex)	Data (Hex)	Description
W	FB	00	
R	FF	XX	讀取 0xFF 會得到 Vender_ID: 61

4. MTP FLOW AND CONTENT

4.1 Normal MTP Flow



4.1.1 External VMTP Flow

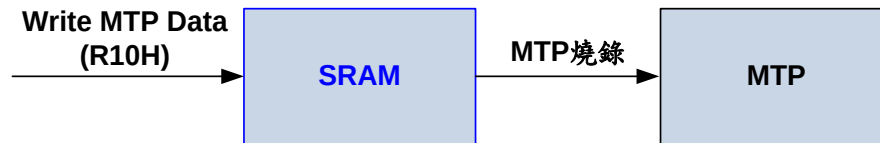
Addr.(Hex)	Start Addr.	Counts	R/W	Data(Hex)	Comment
1.HW reset 2.Check Busy_N = "H"					
A2	1	5	W	0X	VMTP=1 PGM_SADDR[15:0] will be set 0x0000, PGM_DSIZE[15:0] will be set 0x1800.
90	0	0	W	-	PG MODE
10	1	6144	W	XX	寫入欲燒入的 MTP data
VMTP/VPP 接 10.1V					
91	0	0	W	-	APG
1.Delay 60ms 2.Check Busy_N = "H" 3.VMTP/VPP off					
MTP Read 4.HW reset 5.Check Busy_N = "H"					
92	1	6145	R	XX	檢查 MTP 是否有燒錄成功： - P0 為 dummy - Bank 0: P1~P6020 為燒入的 data (P6021~P6144 為 IC 使用，可忽略)

4.1.2 Internal VMTP Flow

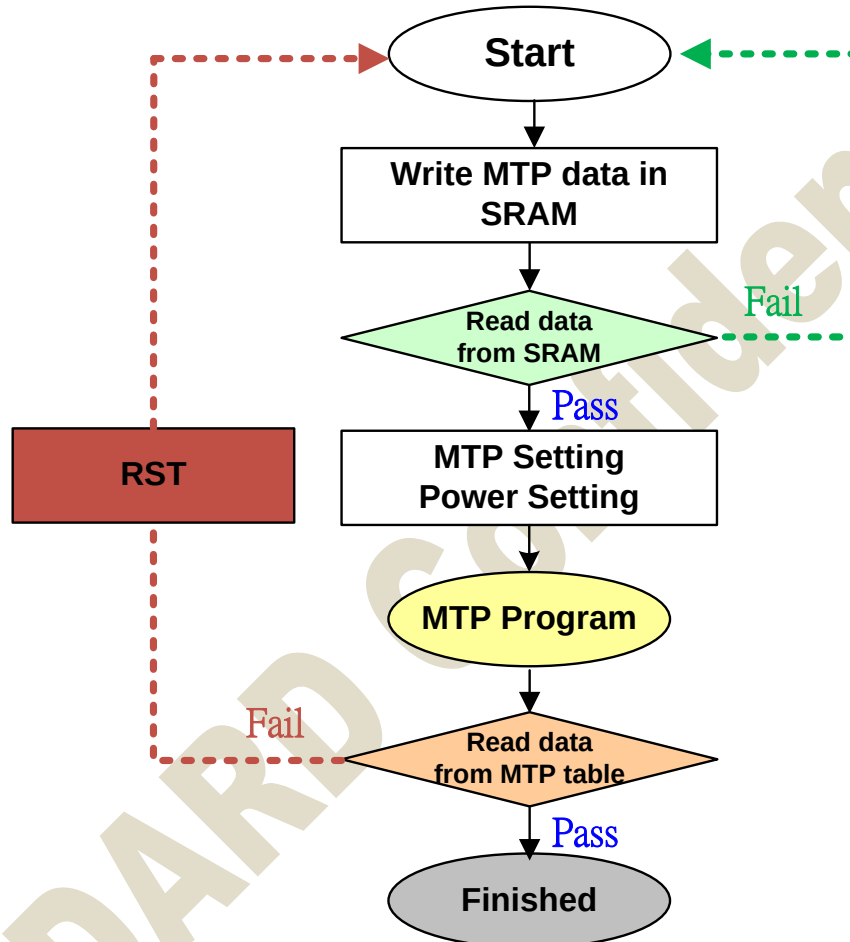
Addr.(Hex)	Start Addr.	Counts	R/W	Data(Hex)	Comment
1.HW reset 2.Check Busy_N = "H"					
4D	1	1	W	78	Power setting
00	1	2	W	2F,89	
06	1	7	W	05, 00, 3F,0A,25,10,1A	
01	1	5	W	0F,00,47,47,47,47	Setting VMTP voltage
A2	1	5	W	10,00,00,01,80	Setting internal VMTP Setting MTP Program size
04	0	0	W		Power-on
1. Check Busy_N = "H" 2. Delay 50ms					
90	0	0	W	-	PG MODE
10	1	6144	W	XX	寫入欲燒入的 MTP data
91	0	0	W	-	APG
1.Delay 60ms 2.Check Busy_N = "H"					
MTP Read 3.HW reset 4.Check Busy_N = "H"					
92	1	6145	R	XX	檢查 MTP 是否有燒錄成功： - P0 為 dummy - Bank 0: P1~P6020 為燒入的 data (P6021~P6144 為 IC 使用，可忽略)

4.2 MTP Flow with SRAM Detection

A. Block digram



B. Flow



4.2.2 Default Setting Format in MTP

	Addr. (Dec)	Addr. (Hex)	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	Value (Hex)	
R9FH	5879-5978	16F7-175A	User Reserved bytes									FF
--	5979	175B	Enable MTP Setting (0xA5)									A5
R00H	5980	175C	RES[1:0]		PST_MODE	-	UD	SHL	SHD_N	RST_N	0F	
	5981	176D	LUT_EN	-	FOPT	VCMZ	TS_AUTO	TIEG	NORG	VC_LUTZ	09	
R01H	5982	175E	-	-	-	-	V_MODE	VSC_EN	VDS_EN	VDG_EN	07	
	5983	175F	-	-	-	-	-	-	VGP[1:0]		00	
	5984	1760	-	VSPL_0[6:0]								00
	5985	1761	-	VSP_1[6:0]								00
	5986	1762	-	VSN_1[6:0]								00
	5987	1763	-	VSPL_1[6:0]								00
R02H	5988	1764	-	-	-	-	-	-	-	EDSE	00	
R03H	5989	1765	-		T_VDPG_OFF [1:0]		-		T_VDS_OFF [1:0]		00	
	5990	1766	VGP_LEN[3:0]				VGP_EXT[3:0]				54	
	5991	1767	XON_DLY[3:0]				XON_LEN[3:0]				44	
R06H	5992	1768	-	-	-	-	PHB_SFT[1:0]		PHA_SFT[1:0]		00	
	5993	1769	-	-	PHA_ON[5:0]						06	
	5994	176A	-	-	PHA_OFF[5:0]						02	
	5995	176B	-	-	PHB_ON[5:0]						07	
	5996	176C	-	-	PHB_OFF[5:0]						02	
	5997	176D	-	-	PHC_ON[5:0]						07	
	5998	176E	-	-	PHC_OFF[5:0]						02	
R12H	5999	177F	-	-	-	-	-	-	-	AC/DC VCOM	00	
R30H	6000	1770	-	-	-	-	Dyna	FR[2:0]		02		
R50h	6001	1771	VBD[2:0]			DDX	CDI[3:0]				97	
-	6002	1772	Reserved									02
	6003	1773	Reserved									02
R61H	6004	1774	-	-	-	-	-	-	HRES[9]	HRES[8]	00	
	6005	1775	HRES[7:3]						0	0	00	
	6006	1776	-	-	-	-	-	-	VRES[9]	VRES[8]	00	
	6007	1777	VRES[7:0]									00
R65H	6008	1778	-	-	-	-	-	-	S_start(9)	S_start(8)	00	
	6009	1779	S_start(7)	S_start(6)	S_start(5)	S_start(4)	S_start(3)	S_start(2)	0	0	00	
	6010	177A	-	-	-	-	-	-	G_start(9)	G_start(8)	00	
	6011	177B	G_start(7)	G_start(6)	G_start(5)	G_start(4)	G_start(3)	G_start(2)	G_start(1)	G_start(0)	00	
-	6012	177C	Reserved									FF
	6013	177D	Reserved									FF
	6014	177E	Reserved									FF
	6015	177F	Reserved									00
RE3H	6016	1780	VCOM_W[3:0]				SD_W[3:0]				00	
RE4H	6017	1781	-	-	-	-	-	-	LVD_SEL[1:0]		03	
-	6018	1782	Reserved									03
	6019	1783	Reserved									1C
	6020	1784	Reserved									00
--	6021-6143	1785-17FF	JD setting									FF

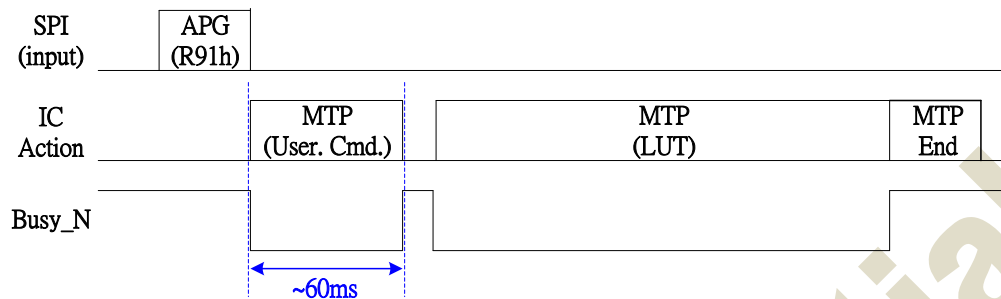
Note:

1.Default setting 內 0xB20 MTP data 需設定為 A5

3.上表 value 請參考，需依實際 initial code 為主來填入

4.3 BUSY_N flag of MTP Program

RA1h 燒錄的期間，MTP 會分成兩部份(user cmd. & LUT)去燒錄，所以 busy_n 會分兩次拉 low，建議下 APG(RA1h) cmd.後，**delay 60ms** 再做 busy_n 拉 high 的偵測，來確定是否燒錄完成。



5. REVISION HISTORY

Revision	Content	Date
1.0	New Issue	2022/10/20
1.1	Modify initial code	2022/12/26

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